

Úloha 3.2

Označme $\mathbf{v}_1 = 1$, $\mathbf{v}_2 = x$ a $\mathbf{v}_3 = x^2$.

(a) Označme vektory hledané báze jako $\mathbf{u}_1, \mathbf{u}_2, \mathbf{u}_3$. Pak je

$$\mathbf{u}_1 = \frac{1}{\|\mathbf{1}\|} = \frac{1}{\sqrt{\int_1^5 1 \cdot 1 \, dx}} = \frac{1}{2}.$$

Dále

$$\mathbf{u}'_2 = \mathbf{v}_2 - \langle \mathbf{u}_1, \mathbf{v}_2 \rangle \mathbf{u}_1 = x - \frac{1}{2} \int_1^5 \frac{1}{2} x \, dx = x - 3,$$

dopočítáme tedy $\mathbf{u}_2$ normalizací:

$$\mathbf{u}_2 = \frac{x - 3}{\|x - 3\|} = \frac{x - 3}{\sqrt{\int_1^5 (x - 3)^2 \, dx}} = \frac{\sqrt{3}}{4} (x - 3).$$

Obdobně pro třetí vektor:

$$\begin{aligned} \mathbf{u}'_3 &= \mathbf{v}_3 - \langle \mathbf{u}_1, \mathbf{v}_3 \rangle \mathbf{u}_1 - \langle \mathbf{u}_2, \mathbf{v}_3 \rangle \mathbf{u}_2 \\ &= x^2 - \frac{1}{2} \int_1^5 \frac{1}{2} x^2 \, dx - \frac{\sqrt{3}}{4} (x - 3) \int_1^5 \frac{\sqrt{3}}{4} (x - 3) x^2 \, dx \\ &= x^2 - 6(x - 3) - \frac{31}{3} \end{aligned}$$

a znormalizujeme na

$$\begin{aligned} \mathbf{u}_3 &= \frac{x^2 - 6(x - 3) - \frac{31}{3}}{\|x^2 - 6(x - 3) - \frac{31}{3}\|} = \frac{x^2 - 6(x - 3) - \frac{31}{3}}{\sqrt{\left[\int_1^5 \left(x^2 - 6(x - 3) - \frac{31}{3} \right)^2 \, dx \right]}} \\ &= \frac{3\sqrt{5}}{16} \left(x^2 - 6(x - 3) - \frac{31}{3} \right). \end{aligned}$$

Nalezená ortonormální báze je tedy

$$B = \left(\frac{1}{2}, \frac{\sqrt{3}}{4} (x - 3), \frac{3\sqrt{5}}{16} \left(x^2 - 6(x - 3) - \frac{31}{3} \right) \right).$$

(b) Protože je B ortonormální báze prostoru

$$\text{Span} \{1, x, x^2\},$$

je podle věty 8.54 ze skript ortogonální projekcí vektoru $\sin x$ do tohoto prostoru vektor

$$\begin{aligned} \mathbf{w} &= \left\langle \frac{1}{2}, \sin x \right\rangle \frac{1}{2} + \left\langle \frac{\sqrt{3}}{4}(x-3), \sin x \right\rangle \frac{\sqrt{3}}{4}(x-3) \\ &+ \left\langle \frac{3\sqrt{5}}{16} \left(x^2 - 6(x-3) - \frac{31}{3} \right), \sin x \right\rangle \frac{3\sqrt{5}}{16} \left(x^2 - 6(x-3) - \frac{31}{3} \right). \end{aligned}$$

Dopočteme:

$$\begin{aligned} &= \frac{1}{2} \int_1^5 \frac{\sin x}{2} dx + \frac{\sqrt{3}}{4}(x-3) \int_1^5 \frac{\sqrt{3} \sin x}{4}(x-3) dx \\ &+ \frac{3\sqrt{5}}{16} \left(x^2 - 6(x-3) - \frac{31}{3} \right) \int_1^5 \frac{3\sqrt{5}}{16} \left(x^2 - 6(x-3) - \frac{31}{3} \right) \sin x dx \\ &\approx -0,074x^2 - 1,179x + 2,774. \end{aligned}$$

(c) Norma rozdílu $\sin x - (-0,074x^2 - 1,179x + 2,774)$ je rovna

$$\begin{aligned} \|\sin x + 0,074x^2 + 1,179x - 2,774\| &= \sqrt{\int_1^5 (\sin x + 0,074x^2 + 1,179x - 2,774)^2 dx} \\ &\approx 3,915. \end{aligned}$$